



# HEALTHCARE BILLING VARIANCE ANALYSIS



Investigating why patients with the same medical condition are billed different amounts



## BUSINESS QUESTION

Why are patients with the same medical condition billed different amounts?



## DATASET

55,501 patient billing records after data cleaning



## APPROACH

Exploratory data analysis across 6 key factors to identify potential drivers of billing variation

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*Turning data into insights to improve healthcare decisions.* ”

## KEY FACTORS ANALYZED



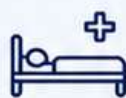
Age &  
Length of Stay



Hospital



Insurance  
Provider



Admission  
Type



Medication



Test  
Results



# Healthcare Billing Variance Analysis

*Investigating why patients with the same medical condition are billed different amounts*



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Tools Used: Microsoft Excel



## About This Project

This project examines why patients with the same medical condition are billed different amounts. Using **55,501 patient billing records**, the analysis explores potential factors influencing billing variation and presents evidence-based findings.



## Business Question

Patients with the same medical condition are billed different amounts. This project investigates whether factors such as **Age, Length of Stay, Hospital, Insurance Provider, Admission Type and Medication** can explain these billing differences.



## Analysis Approach

**To answer the business question, the analysis followed a structured approach:**

- Verified the quality and consistency of the data before beginning the analysis.
- Analyzed numeric variables, including age and length of stay, to identify relationships with billing amounts within each medical condition.
- Compared billing patterns across categorical factors such as hospital, insurance provider, admission type, medication, and test results.
- Combined findings from all analyses to draw evidence-based conclusions rather than relying on a single factor.

## Investigation 1 – Data Quality Assessment

Before beginning the analysis, I assessed the quality of the dataset to ensure it was reliable and suitable for answering the business question.

| Check                                    | Result               |
|------------------------------------------|----------------------|
| Missing values                           | 0                    |
| Duplicate rows                           | 118 rows             |
| Negative Billing Amount (impossible)     | 107 rows             |
| Billing Amount range                     | -\$2,008 to \$52,764 |
| <b>Final dataset size after cleaning</b> | <b>55276 rows</b>    |

### Key Insight

The dataset had no missing values, but contained **118 duplicate records** and **106 records with negative billing amounts** — both invalid for analysis. These **225 rows were removed**, reducing the dataset from 55,500 to **55275 records** used in all subsequent investigations.

## Investigation 2 – Do Age and Length of Stay Explain Billing Differences?

To determine whether patient characteristics influence billing amounts, I analyzed the relationship between **Age**, **Length of Stay**, and **Billing Amount** within each medical condition using correlation analysis.

| Medical Condition | Age Correlation | Length of Stay Correlation | Interpretation             |
|-------------------|-----------------|----------------------------|----------------------------|
| Cancer            | -0.005          | -0.001                     | No meaningful Relationship |
| Obesity           | -0.016          | 0.000                      | No meaningful Relationship |
| Diabetes          | 0.005           | -0.002                     | No meaningful Relationship |
| Asthma            | -0.014          | -0.019                     | No meaningful Relationship |
| Hypertension      | -0.005          | -0.002                     | No meaningful Relationship |
| Arthritis         | 0.012           | -0.008                     | No meaningful Relationship |

### Key Insight

The correlation values for **Age** and **Length of Stay** were consistently close to **zero** across all medical conditions. This indicates **no meaningful relationship** between these variables and Billing Amount. Based on the available data, neither patient age nor length of stay explains why patients with the same medical condition receive different billing amounts.

## Investigation 3 – Does Hospital Explain Billing Differences?

To determine whether the hospital influences billing amounts, I compared the **average billing amount** across hospitals within each medical condition.

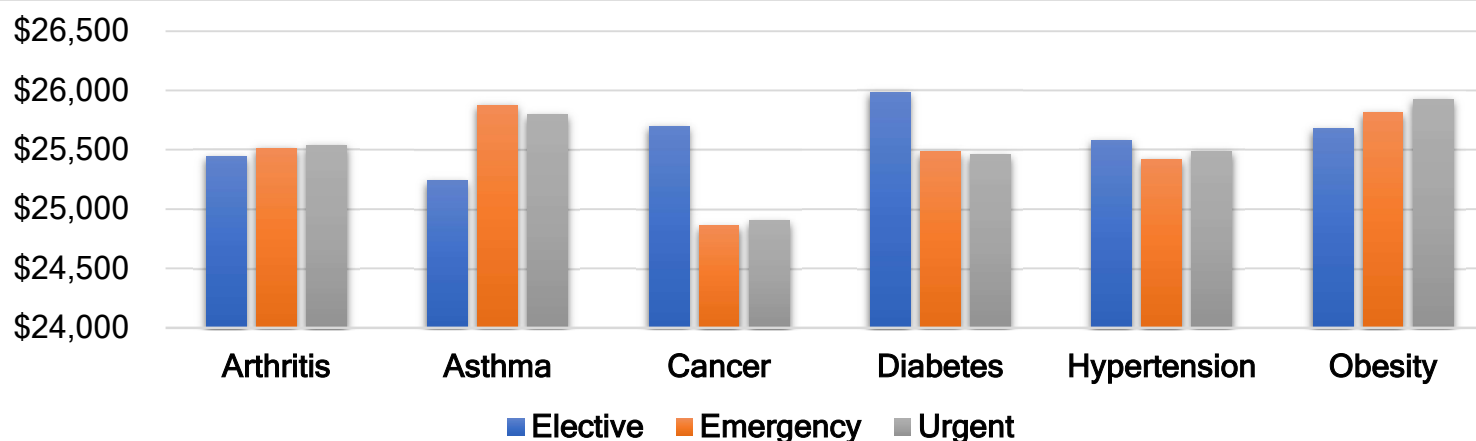
| Average of Billing Amount     | Medical Condition |           |           |           |              |           |
|-------------------------------|-------------------|-----------|-----------|-----------|--------------|-----------|
| Medical Condition Vs Hospital | Arthritis         | Asthma    | Cancer    | Diabetes  | Hypertension | Obesity   |
| Lakeside Medical Center       | \$ 25,257         | \$ 25,629 | \$ 24,961 | \$ 25,692 | \$ 25,551    | \$ 25,577 |
| Mercy Grace Hospital          | \$ 25,431         | \$ 25,923 | \$ 24,986 | \$ 25,892 | \$ 25,796    | \$ 25,930 |
| Riverside Community Hospital  | \$ 25,608         | \$ 26,006 | \$ 25,461 | \$ 25,312 | \$ 25,378    | \$ 25,908 |
| St. Anthony's Hospital        | \$ 25,450         | \$ 25,577 | \$ 25,404 | \$ 26,125 | \$ 25,404    | \$ 25,541 |
| Sunrise General Hospital      | \$ 25,749         | \$ 25,023 | \$ 24,997 | \$ 25,149 | \$ 25,340    | \$ 26,068 |

### Key Insight

The average billing amount remained close to **\$25,500** across all hospitals and medical conditions. Although small differences were observed between hospitals, the variation was not consistent enough to indicate that hospital location is a major driver of billing differences. Based on the available data, **hospital does not appear to explain why patients with the same medical condition are billed different amounts.**

## Investigation 4 – Does Admission Type Explain Billing Differences?

To determine whether **Admission Type** influences billing amounts, I compared the **average billing amount** across **Elective**, **Emergency**, and **Urgent** admissions within each medical condition.

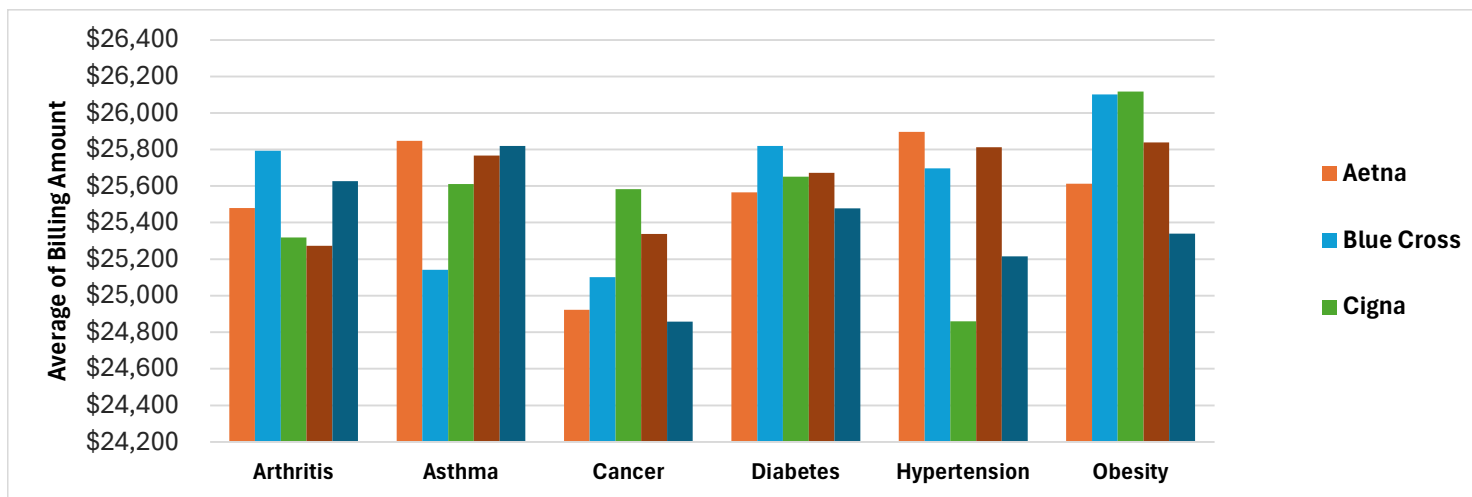


### Key Insight

Average billing amounts were broadly similar across **Elective**, **Emergency**, and **Urgent** admissions for each medical condition. Although small differences were observed, the variation was not consistent across conditions. Based on the available data, **Admission Type does not appear to explain why patients with the same medical condition are billed different amounts.**

## Investigation 5 – Does Insurance Provider Explain Billing Differences?

To determine whether Insurance Provider influences billing amounts, I compared the average billing amount across **insurance providers** within each **medical condition**.

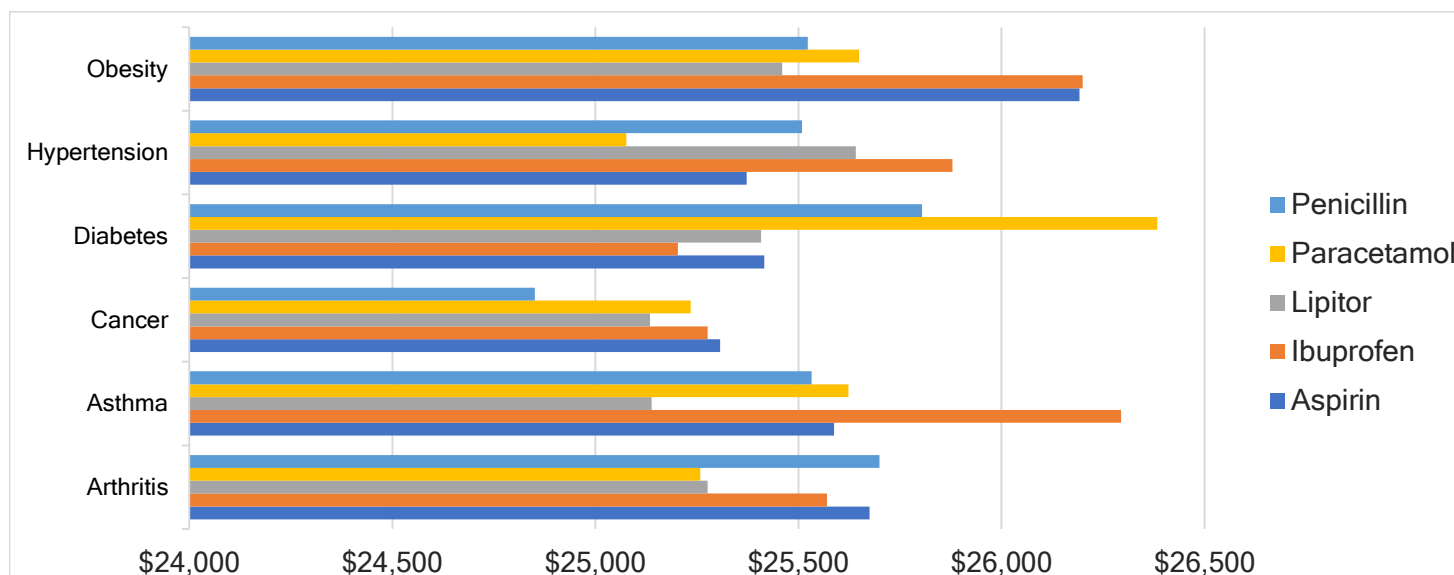


### Key Insight

Average billing amounts were similar across all insurance providers. Although small differences were observed, **no consistent pattern indicates that insurance provider explains why patients with the same medical condition are billed different amounts.**

## Investigation 6 – Does Medication Explain Billing Differences?

To determine whether **Medication** influences billing amounts, I compared the **average billing amount** across different medications within each medical condition.





## Key Insight

Average billing amounts varied slightly across different medications within each medical condition. However, **no consistent pattern** was observed across conditions. Based on the available data, **Medication does not appear to explain why patients with the same medical condition are billed different amounts.**

## Conclusion

This analysis examined whether **Age, Length of Stay, Hospital, Insurance Provider, Admission Type, and Medication** explain why patients with the same medical condition receive different bills. Before analysis, **118 duplicate records** and **107 invalid negative billing amounts** were identified and removed to ensure reliable results.

None of the tested variables showed a meaningful relationship with Billing Amount. Correlations for Age and Length of Stay were **near zero**, and average billing remained tightly clustered between **\$25,000–\$26,100** across every hospital, insurance provider, admission type, and medication category — regardless of which group was examined.

This **unusually consistent range across all variables** is itself a notable finding: it suggests the dataset may not fully capture real-world healthcare billing behavior, where cost drivers are typically far more variable. Based on the evidence gathered, **billing differences remain unexplained by the variables available in this dataset.**

## Recommendations

- ❖ None of the factors I tested explained the billing differences, so future analysis could look at things like procedure type, room type, or physician fees.
- ❖ Billing amounts stayed in a similar range across every hospital, insurance provider, and condition — which seems unusual for real healthcare data. Worth checking with the data source to confirm accuracy.
- ❖ This analysis relied on comparing averages and charts. A statistical test like ANOVA would give a more solid answer on whether the differences are real or just random noise.
- ❖ Adding basic data validation at collection (like blocking negative billing amounts) would prevent this kind of cleanup in future datasets.